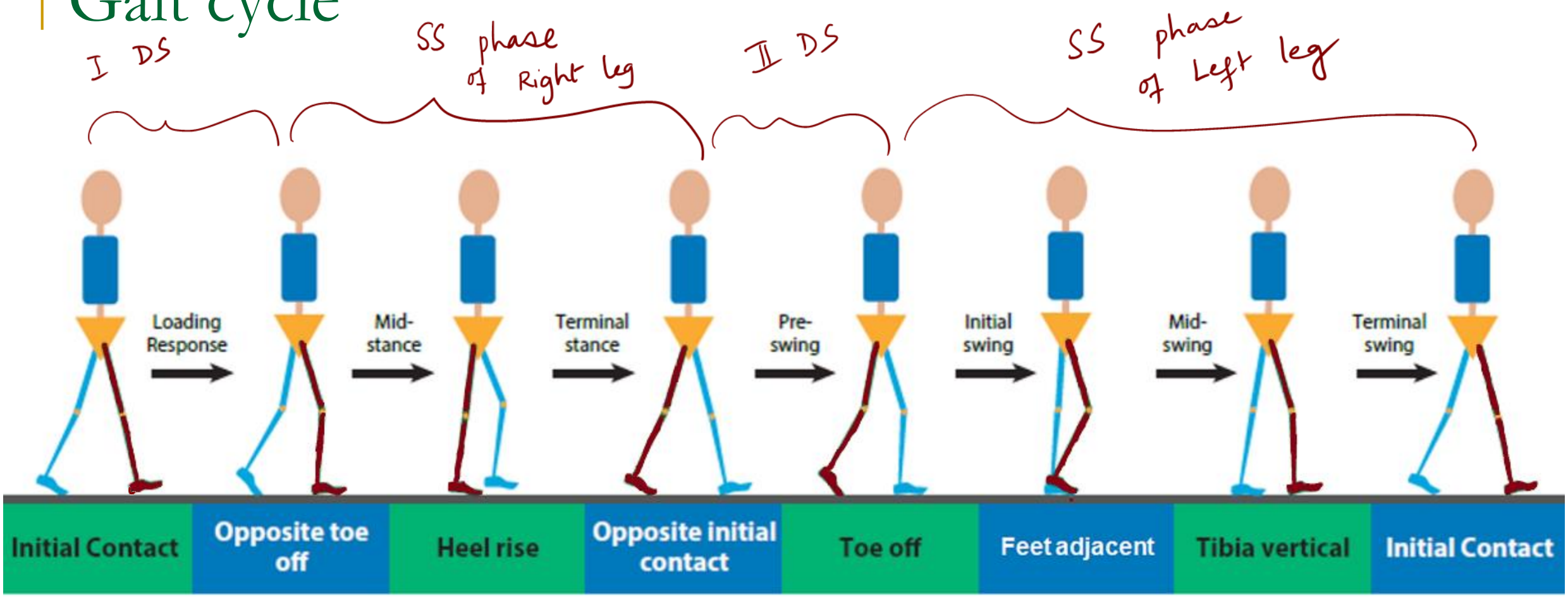
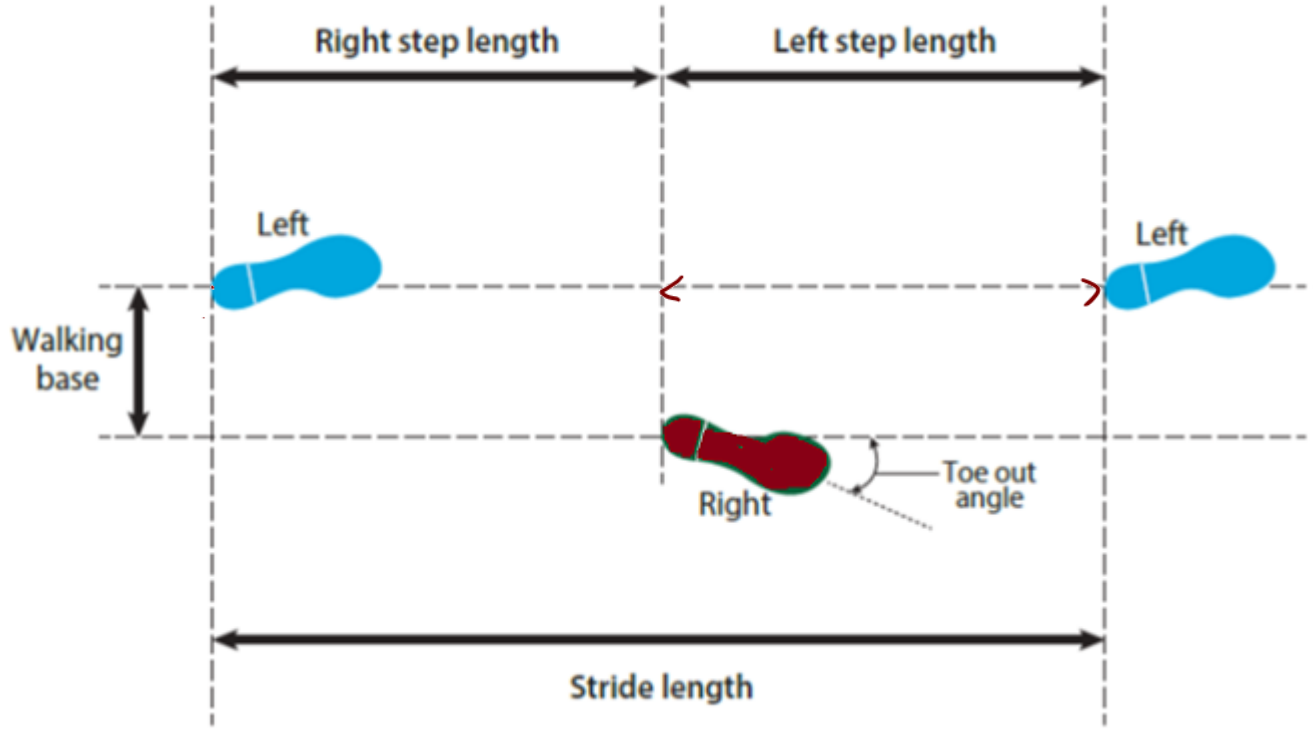
Gait cycle



Each leg is in stance – 60% of the gait cycle

in swing – 40% of the gait cycle

Gait cycle



If there is a shorter step length with one leg, it is likely that the problem is with the other leg.

Symmetric gait - two equal steps

Gait terminology

Cadence: Steps per minute

Cycle time (s) = 120/cadence

Average speed is distance covered per unit time

speed (m/s)=stride length (m)*cadence (steps/min)/120

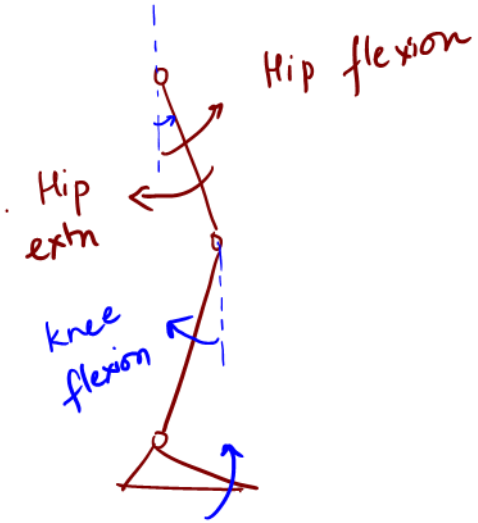
or

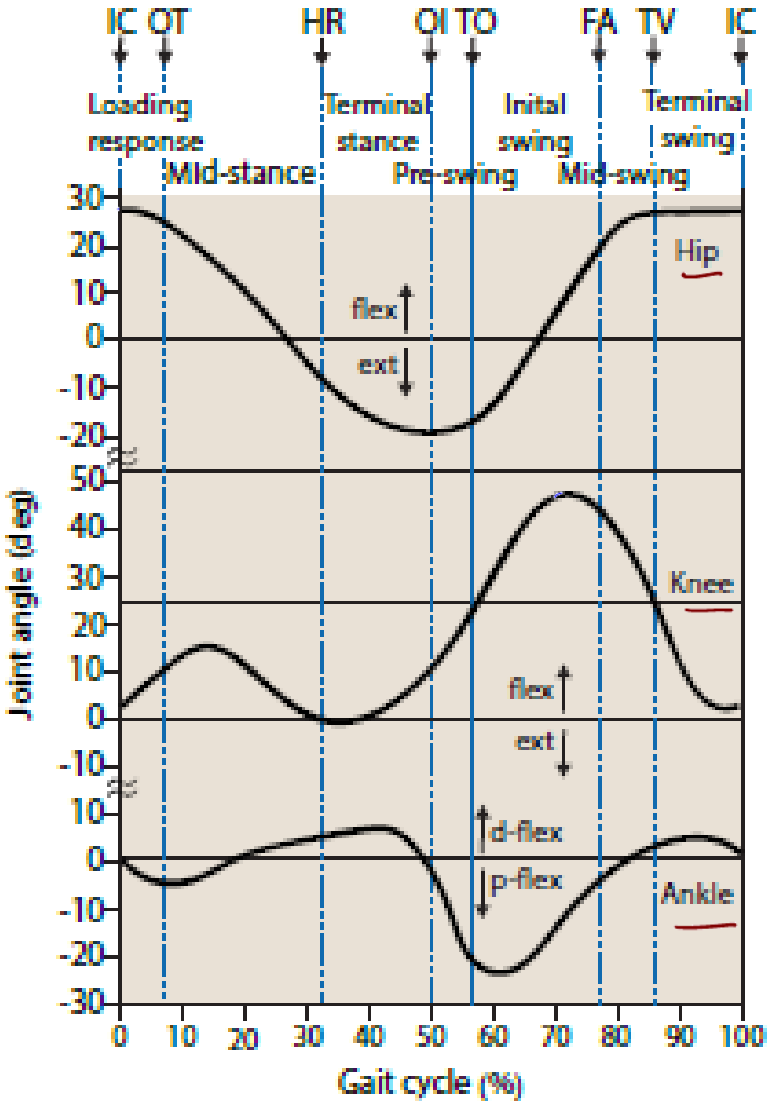
speed (m/s)=stride length (m)/cycle time(s)

1.3 - 1.4 m/s

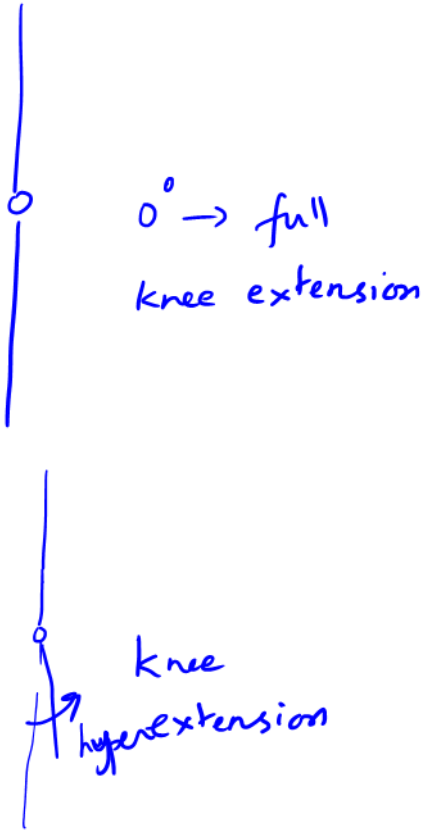
average self-selected walking speed

Joint motions



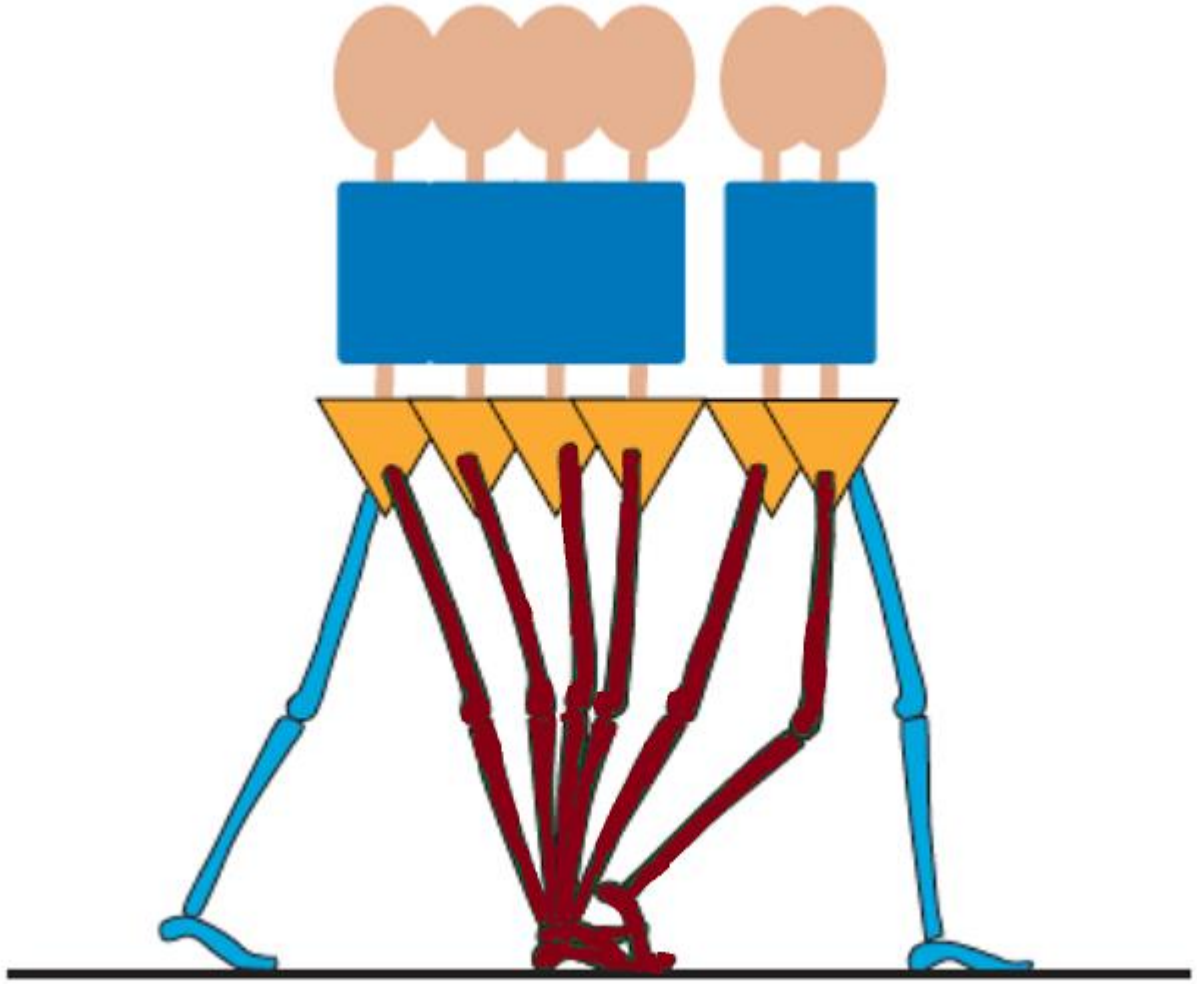


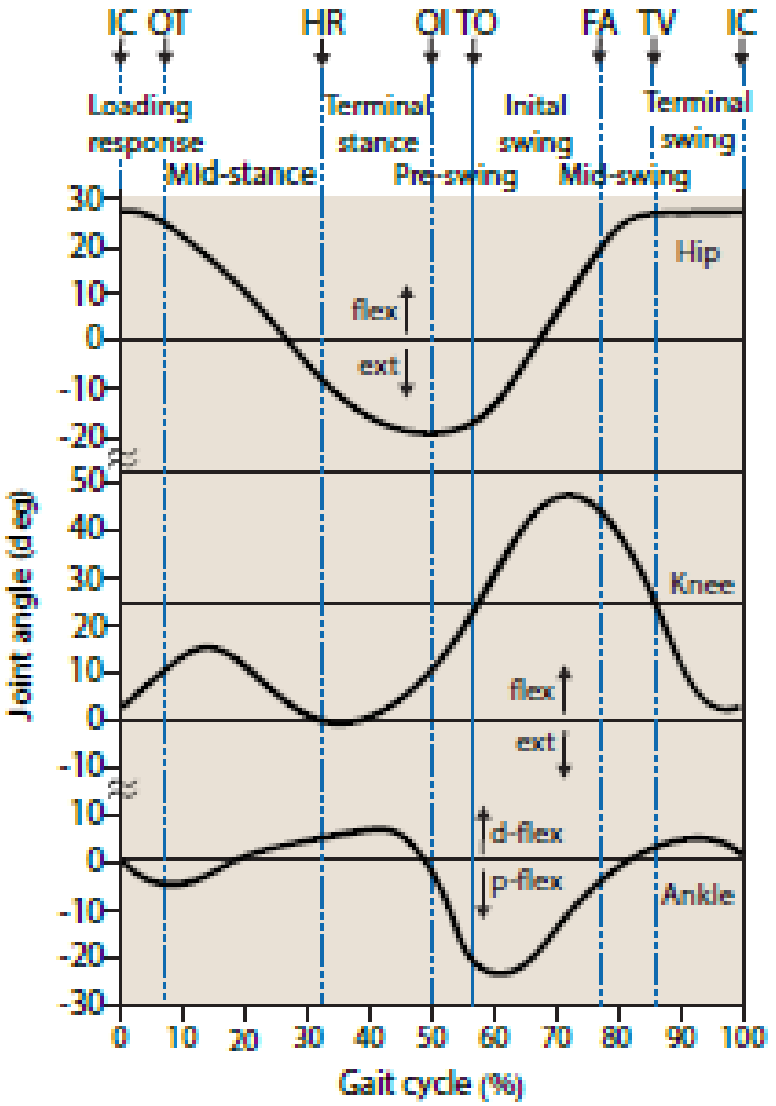




Joint motions: Whittle, MW: *Gait Analysis*

Stance limb during gait





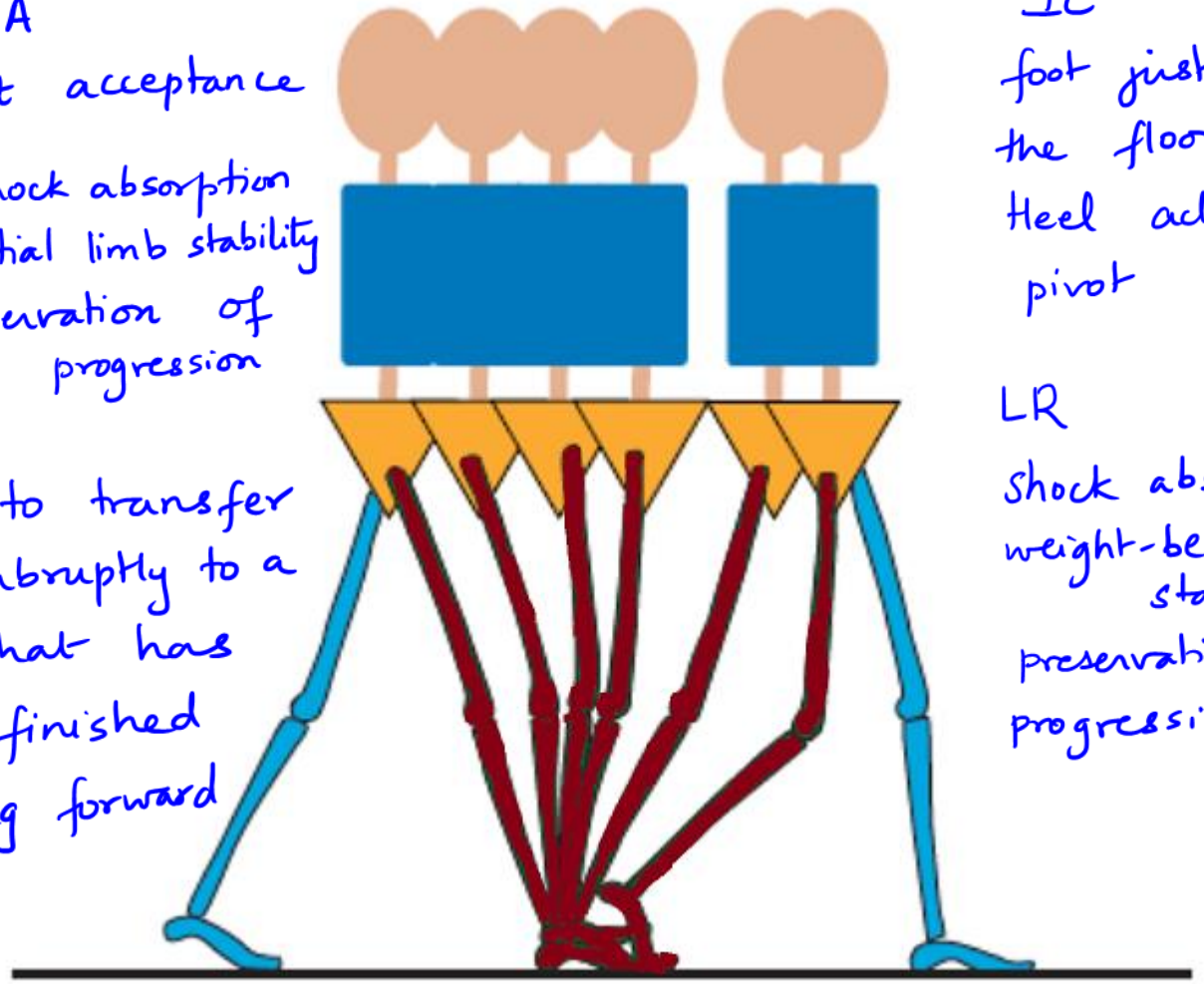
Stance limb during gait

Task A

Weight acceptance

↳ Shock absorption
Initial limb stability
Preservation of progression

Need to transfer BW abruptly to a limb that has just finished swinging forward



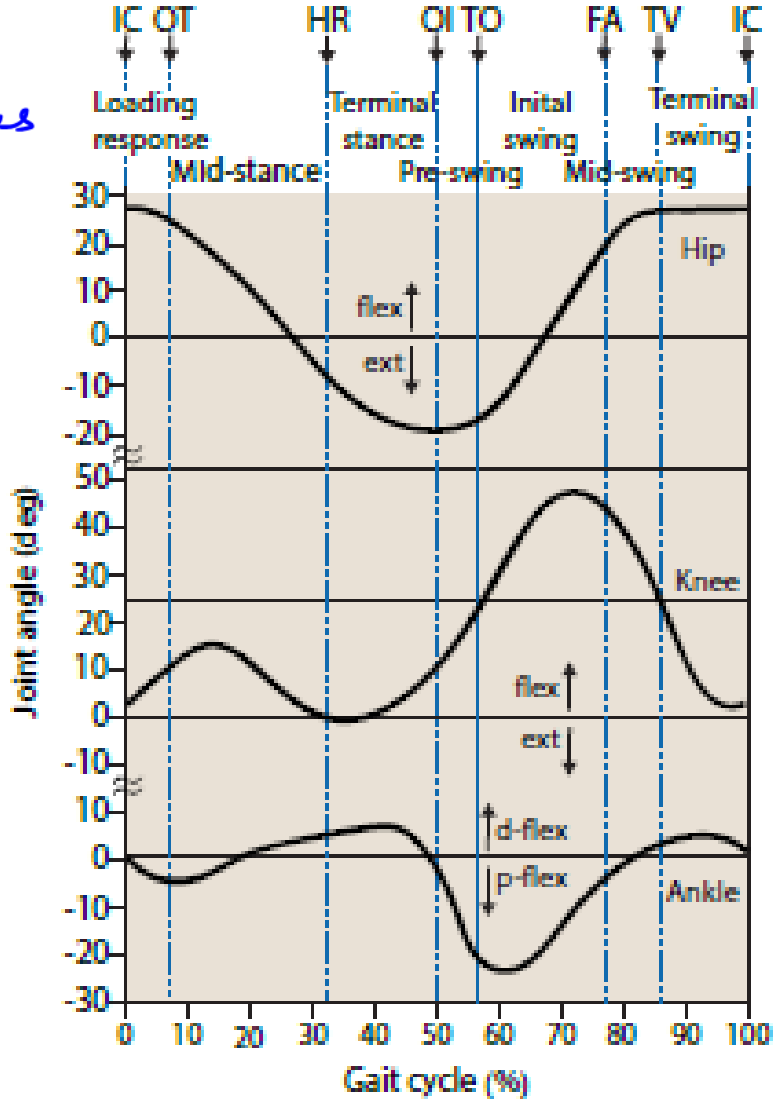
IC

foot just touches the floor

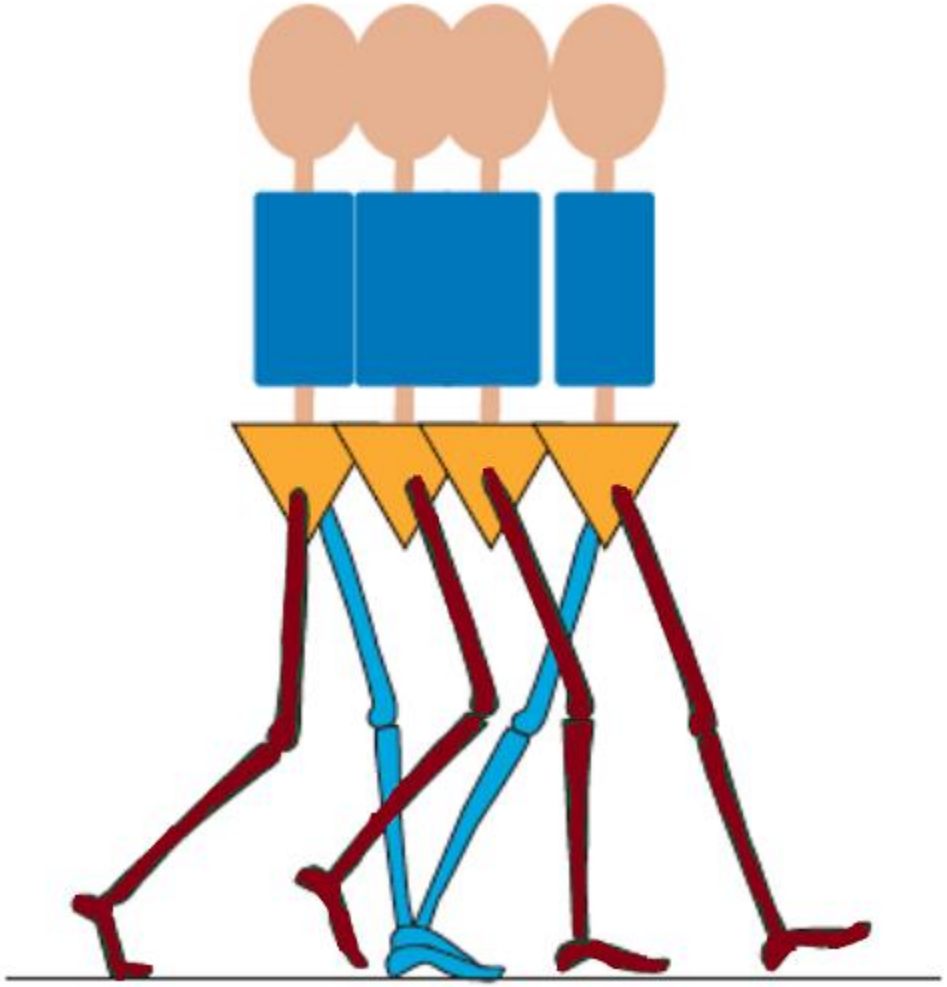
Heel acts as a pivot

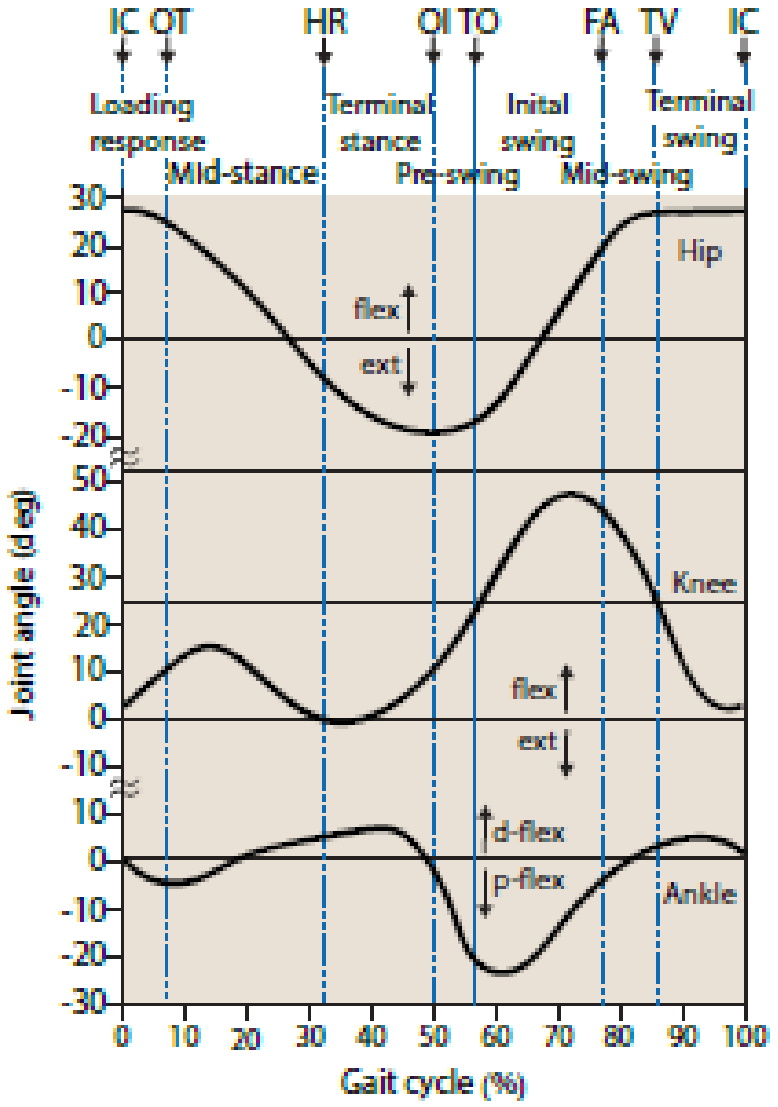
LR

Shock absorption
weight-bearing
stability
preservation of progression

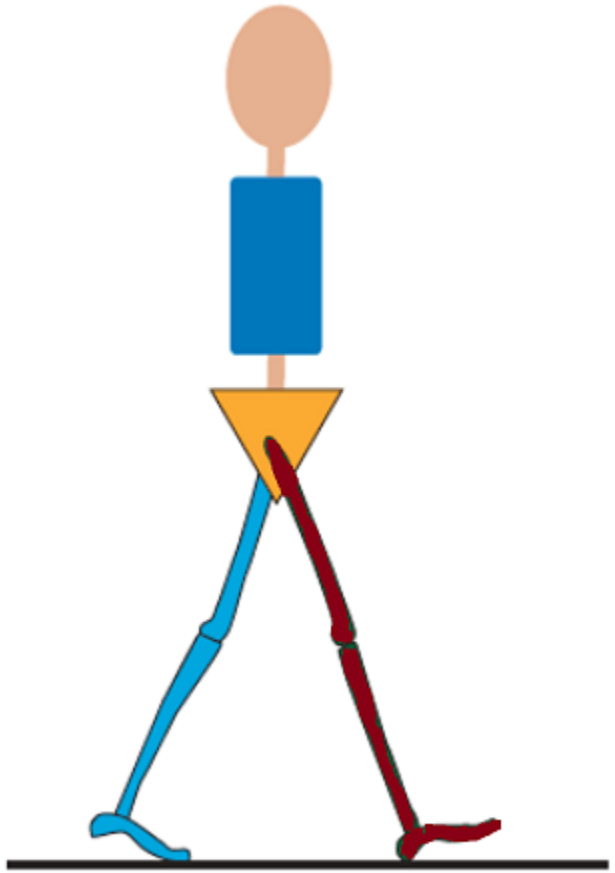


Swing limb during gait

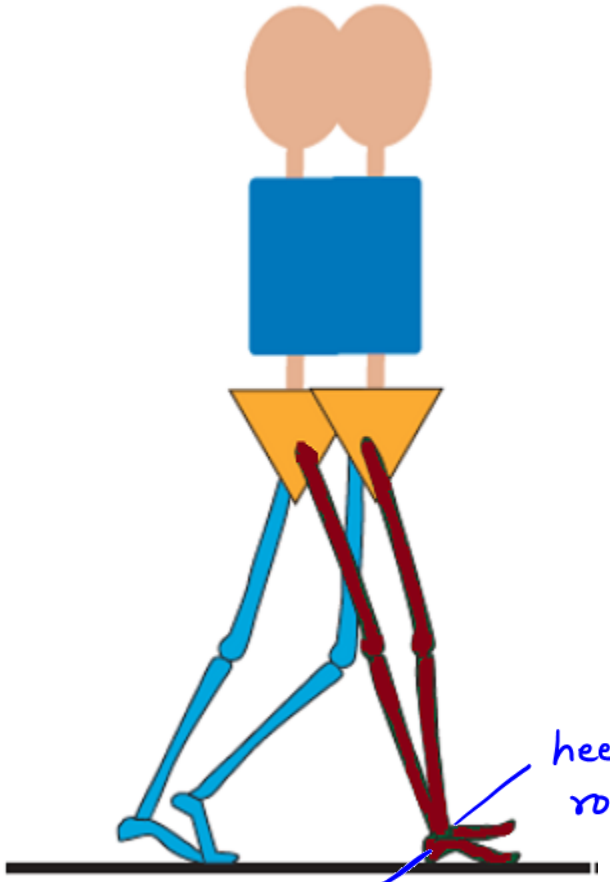




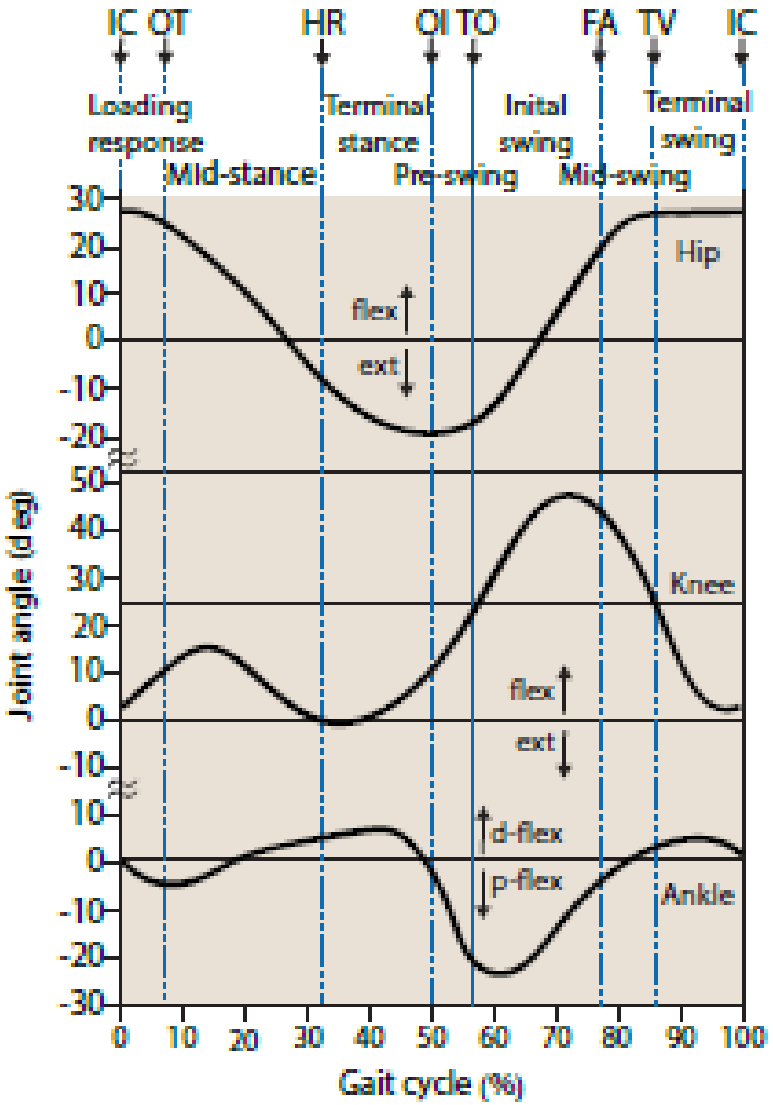
Stance phase



Initial Contact



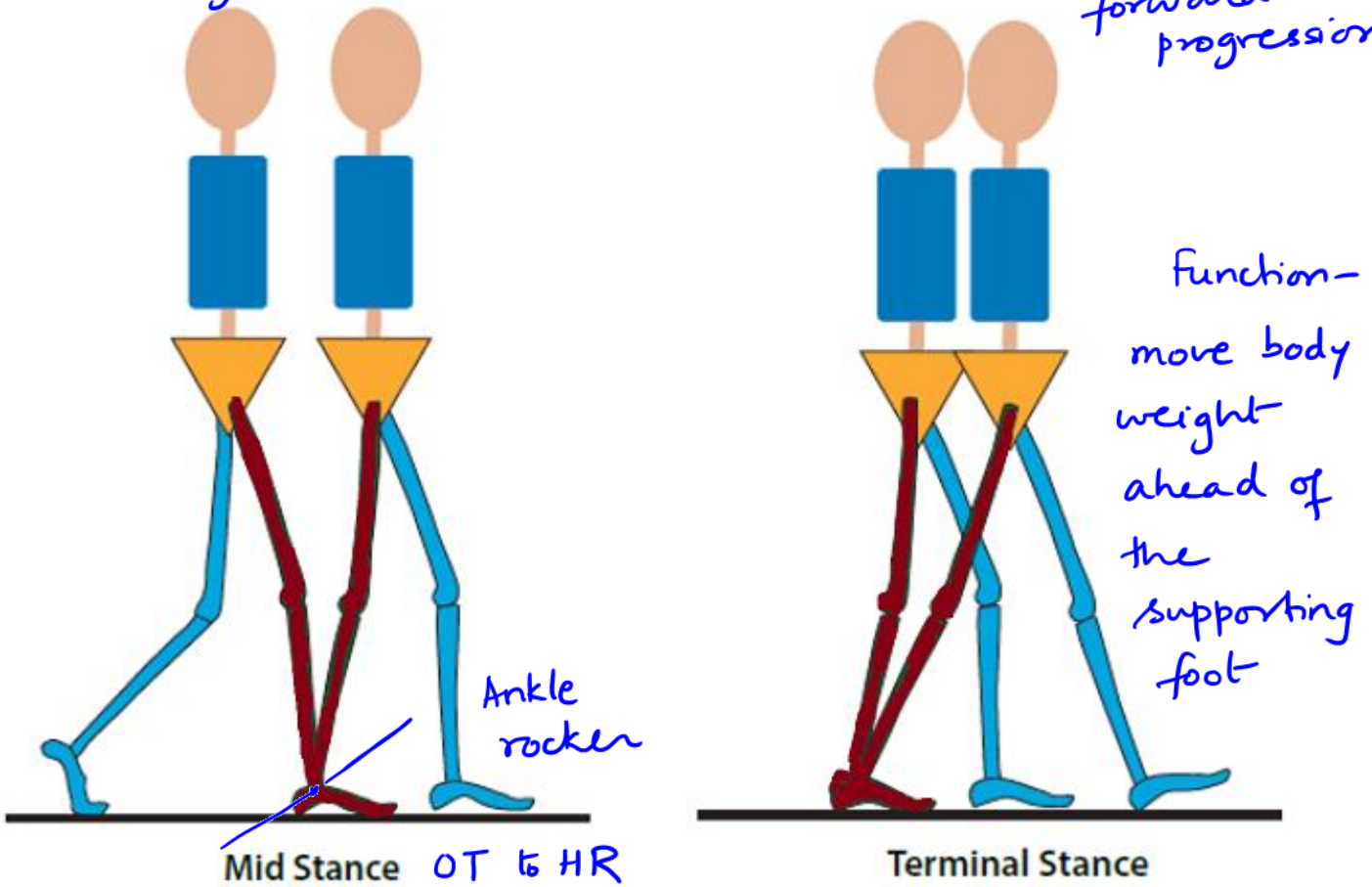
Loading Response



Stance phase

Task B – stability while maintaining progression

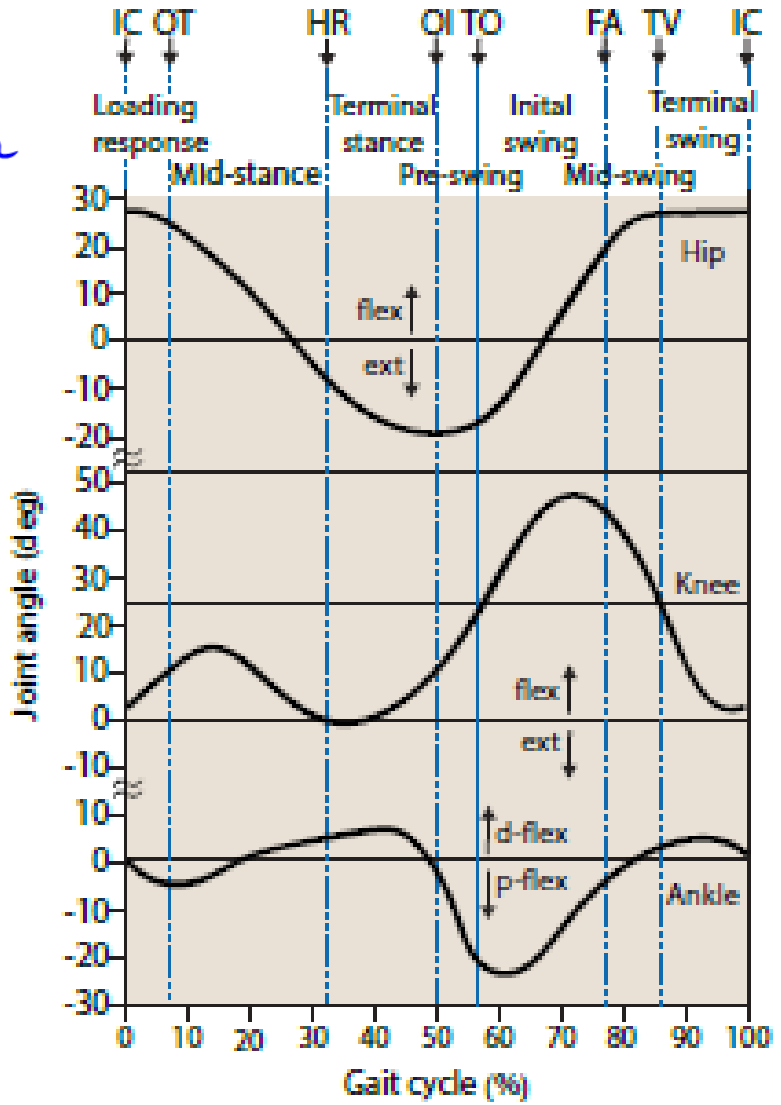
One limb has total responsibility for supporting BW in both sagittal & frontal planes, while continuing forward progression



Function – move body weight ahead of the supporting foot

Mid Stance OT to HR

Progression over stationary foot, stability



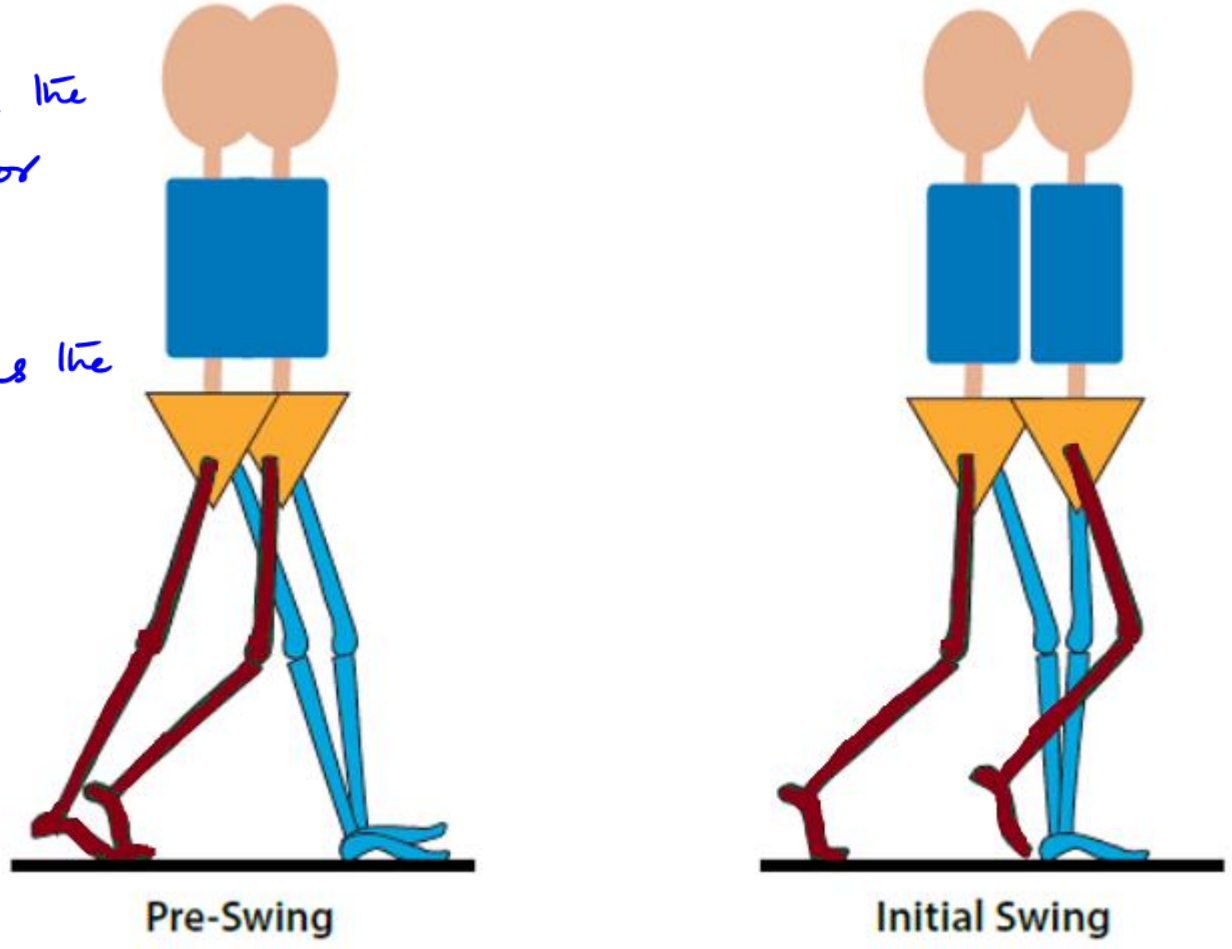
Swing phase

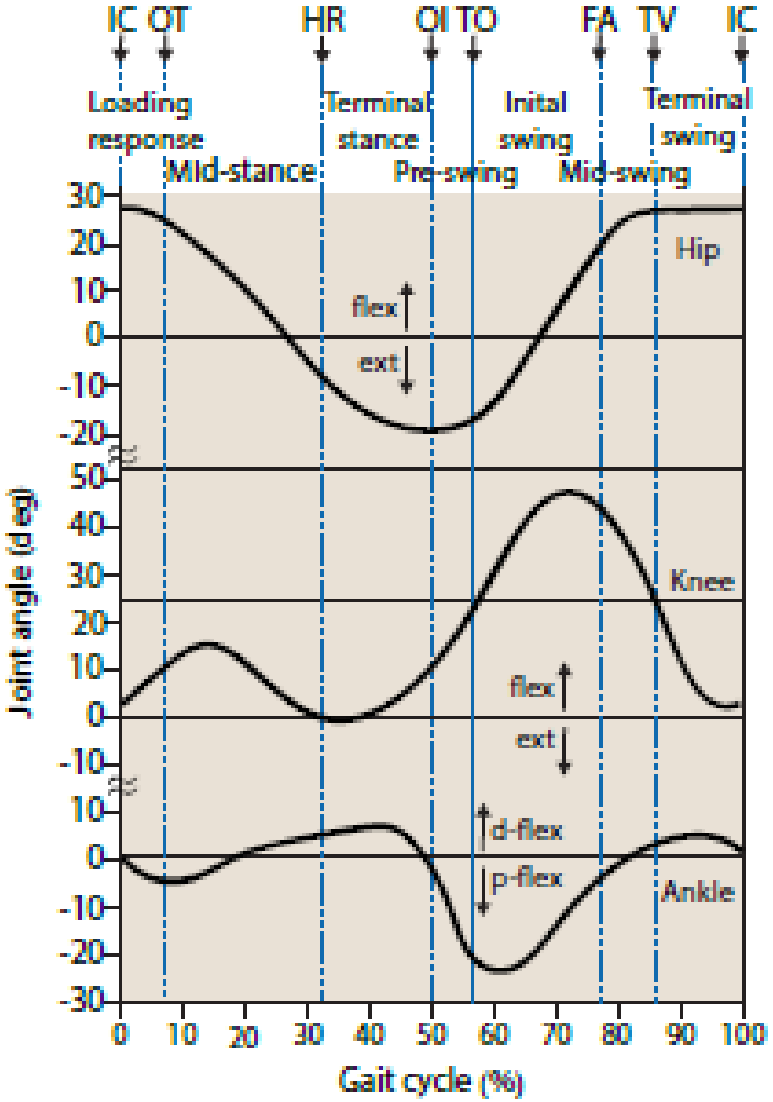
Task C - limb advancement

Obj :

Position the limb for swing

Unloads the limb

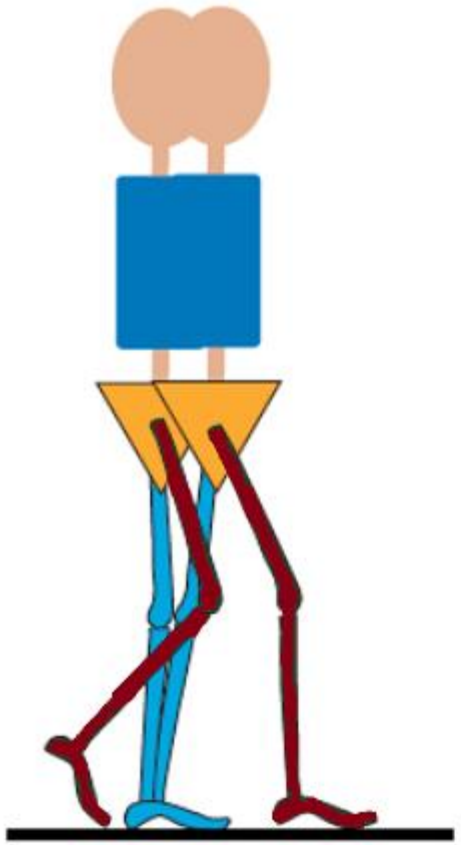




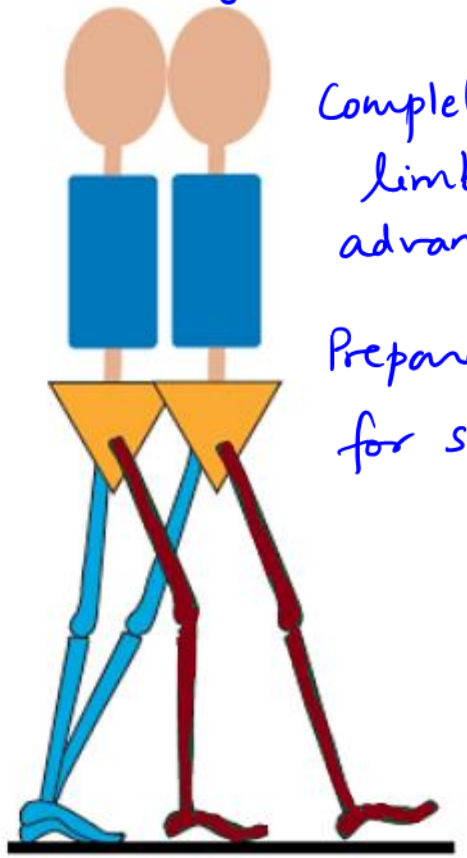
Swing phase

Foot clearance

Shank moves ahead of the thigh



Mid Swing

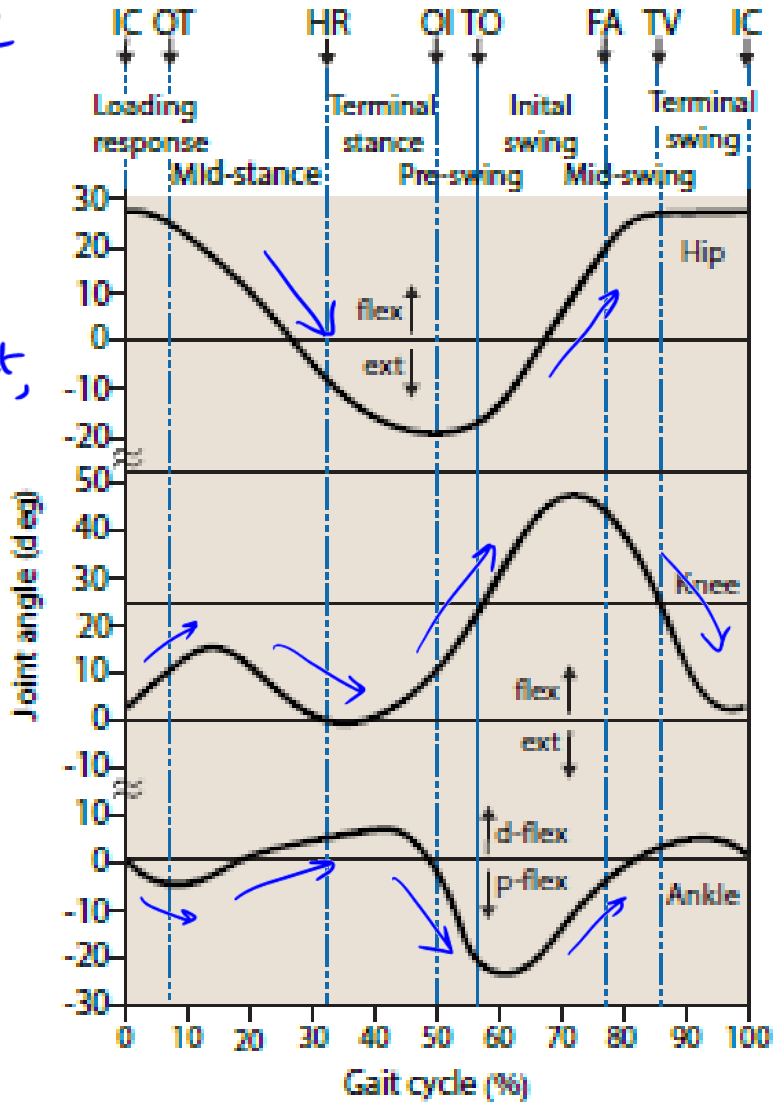


Terminal Swing

Complete limb advancement,

Prepare limb for stance

TV to IC



Action of the foot

